

Leverage, Structural Rank, and Thermodynamic Selection in Information-Processing Systems

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April 15, 2026

Abstract

We study leverage as a structural invariant of finite information-processing systems. For any system A with capabilities $\text{Cap}(A) > 0$, we prove a Five-Way Equivalence:

$$\begin{aligned} \text{DOF}(A) = 1 &\iff \max \text{leverage } L = |\text{Cap}|/\text{DOF} \iff \text{single source of truth} \\ &\iff \text{srnk} = 1 \iff \text{tractable sufficiency} \\ &\iff \text{minimum thermodynamic cost.} \end{aligned}$$

We also prove that the England replication inequality,

$$\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k,$$

reduces to counting: a k -variable system has 2^k states, and the finite inequality $k \leq 2^{k-1}$ yields the entropy gap under Landauer calibration. The thermodynamic bound $E_{\text{decision}} \geq \text{srnk} \cdot k_B T \ln 2$ then follows from the same calibration (BA7).

Within the explicit thermodynamic model used in the paper (Landauer calibration, finite energy, finite signal speed, and nontrivial state space), we derive a physical no-go theorem for polynomial collapse (LP38). The same framework yields unconditional Landauer-linear lower bounds for decision cost and identifies the single-source regime as the unique minimum-cost point. Reliability-style and edit-complexity corollaries are retained only as structural consequences of the same rank parameter.

All theorems are machine-checked in Lean 4 with no `sorry` placeholders. The only explicit physics premises are Landauer calibration, the Second Law (non-negative entropy production), and the Thermodynamic Uncertainty Relation (Barato-Seifert 2015). An assumption ledger records the dependency structure of each physically grounded result.

Keywords: thermodynamics, Landauer principle, entropy production, information theory, structural rank, formal verification, degrees of freedom

1 Introduction

This paper studies finite information-processing systems through a single structural parameter: degrees of freedom. The central result is the Five-Way Equivalence (Theorem 1.1): engineering leverage, epistemic coherence, structural rank, tractable sufficiency, and minimum thermodynamic cost all select the same regime, $\text{DOF} = 1$. The main contribution of this version is the mathematical-physics side of that equivalence: it turns a structural counting parameter into a calibrated entropy and energy law. All claims are verified in Lean 4 with zero `sorry` placeholders.

1.1 Formalization Statistics

Metric	Value
Lines of Lean 4 (local layer)	4122
Theorems/lemmas (local layer)	210
<code>sorry</code> placeholders (local layer)	0
Proof files (local layer)	17
Total imported Lean lines	75943
Total theorem/lemma statements	3301
Total imported <code>sorry</code> placeholders	0
Total proof files in import closure	280
Dependencies (live imports)	
<code>AbstractClassSystem</code>	13849 lines, 677 theorems
<code>Ssot</code>	12507 lines, 591 theorems
<code>DecisionQuotient</code>	45465 lines, 1823 theorems

All proofs build with `lake build`. Axiom dependencies verified via `#print axioms: propext, Quot.sound, Classical.choice`.

1.2 Central Result

Theorem 1.1 (Five-Way Equivalence). *For any architecture A with $\text{Cap}(A) > 0$, the following are equivalent:*

$$\begin{array}{ccccccc}
 \underbrace{\text{DOF}(A) = 1}_{\text{single source}} & \iff & \underbrace{L(A) \text{ maximal}}_{\text{engineering}} & \iff & \underbrace{\text{SSOT}(A)}_{\text{epistemics}} \\
 & \iff & \underbrace{\text{srnk}(A) = 1}_{\text{information}} & \iff & \underbrace{\text{tractable sufficiency}}_{\text{complexity}} & \iff & \underbrace{\text{min thermo cost}}_{\text{physics}}
 \end{array}$$

Proved in Section 5. Machine-checked via `Leverage` $\rightarrow$ `DecisionQuotient` $\rightarrow$ `Ssot` $\rightarrow$ `Mathlib`.

1.3 Contributions

1. **DOF-Structural-Rank Identification (Theorem 5.3):** the canonical decision problem associated with an n -degree-of-freedom system has structural rank n .
2. **Tractability Boundary (Theorem 5.5):** rank-1 sufficiency checking is polynomial-time, while rank greater than one enters the coNP-hard regime in the companion structural-rank theory.
3. **Thermodynamic Selection Theorem (Theorem 5.7):** under Landauer calibration, degree of freedom above one forces strictly higher per-cycle decision cost, and the positive energy floor is unconditional.
4. **England Replication Inequality (Theorem 5.1):** $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$. The proof reduces the entropy gap to counting via $k \leq 2^{k-1}$.¹
5. **Five-Way Convergence (Theorem 1.1):** five independently motivated frameworks select the same structural regime, making the single-source point simultaneously optimal for leverage, coherence, rank, tractability, and thermodynamic cost.

¹ Lean: [L45](#)

What is new. `AbstractClassSystem` and `Ssot` establish two of the five equivalences. This paper closes the physics-facing half of the chain: it identifies structural rank as the information coordinate of DOF, proves thermodynamic selection under Landauer calibration, and derives the England-style replication inequality from finite counting.

1.4 Dependency Chain

1. `AbstractClassSystem`: axis orthogonality $\rightarrow$ error independence (Theorem 3.1)
2. `Ssot`: $\text{DOF} = 1 \iff \text{coherence} \rightarrow \text{second equivalence}$
3. `DecisionQuotient`: tractability boundary and thermodynamic lift $\rightarrow$ fourth and fifth equivalences
4. `Leverage` (this paper): convergence of leverage, rank, tractability, and thermodynamic cost

1.5 Scope

The focus of this manuscript is not software methodology per se, but the mathematical structure linking leverage, structural rank, complexity, and thermodynamic cost. Reliability-style and edit-cost consequences are kept because they expose the same parameter in a more familiar discrete language. A separate software-engineering split develops the architectural decision framework and case-study material in its own venue-facing form.

1.6 Organization

Section 2 defines the structural model. Section 3 derives the reliability-style semantics of DOF. Section 4 records the main leverage consequences needed later. Section 5 proves the convergence theorem and the England inequality. Section 6 situates the paper relative to thermodynamics, information theory, and formalized complexity. Section A describes the Lean mechanization.

2 Foundations

We formalize the core concepts: architecture state spaces, degrees of freedom, capabilities, and leverage.

2.1 Architecture State Space

Definition 2.1 (Architecture). An *architecture* is a tuple $A = (C, S, T, R)$ where:

- C is a finite set of *components* (modules, services, endpoints, etc.)
- $S = \prod_{c \in C} S_c$ is the *state space* (product of component state spaces)
- $T : S \rightarrow \mathcal{P}(S)$ defines valid *transitions* (state changes)
- R is a set of *requirements* the architecture must satisfy

Intuition: An architecture consists of components, each with a state space. The total state space is the product of component spaces. Transitions define how the system can evolve.

Example 2.2 (Microservices Architecture). • $C = \{U, O, P\}$ where $U = \text{UserService}$, $O = \text{OrderService}$, $P = \text{PaymentService}$

- $S_{\text{UserService}} = \text{UserDB} \times \text{Endpoints} \times \text{Config}$
- Similar for other services
- $S = S_{\text{UserService}} \times S_{\text{OrderService}} \times S_{\text{PaymentService}}$

2.2 Degrees of Freedom

Definition 2.3 (Degrees of Freedom). The *degrees of freedom* of architecture $A = (C, S, T, R)$ is a natural number $\text{DOF}(A) \in \mathbb{N}$ counting the independent axes of variation in the state space. Formally, it is the cardinality of a minimal complete orthogonal axis set for S (as established in `AbstractClassSystem`). In the Lean formalization, `Architecture.dof` is a field of type $\mathbb{N}$.

Operational meaning: DOF counts independent modification points. If $\text{DOF}(A) = n$, then n independent changes can be made to the architecture.

Proposition 2.4 (DOF Additivity). *For disjoint architectures $A_1 = (C_1, S_1, T_1, R_1)$ and $A_2 = (C_2, S_2, T_2, R_2)$ with $C_1 \cap C_2 = \emptyset$:*

$$\text{DOF}(A_1 \oplus A_2) = \text{DOF}(A_1) + \text{DOF}(A_2)$$

where $A_1 \oplus A_2 = (C_1 \cup C_2, S_1 \times S_2, T_1 \times T_2, R_1 \cup R_2)$.

²

Proof. Components are disjoint ($C_1 \cap C_2 = \emptyset$), so the axis sets of A_1 and A_2 are disjoint. The axis set of $A_1 \oplus A_2$ is their disjoint union, giving $\text{DOF}(A_1 \oplus A_2) = |\text{axes}(A_1)| + |\text{axes}(A_2)| = \text{DOF}(A_1) + \text{DOF}(A_2)$. ■

Example 2.5 (DOF Calculations). 1. **Monolith:** Single deployment unit $\rightarrow \text{DOF} = 1$

2. n **Microservices:** n independent services $\rightarrow \text{DOF} = n$

3. **Copied Code:** Code duplicated to n locations $\rightarrow \text{DOF} = n$ (each copy independent)

4. **SSOT:** Single source, n derived uses $\rightarrow \text{DOF} = 1$ (only source is independent)

5. k **API Endpoints:** k independent definitions $\rightarrow \text{DOF} = k$

6. m **Config Parameters:** m independent settings $\rightarrow \text{DOF} = m$

2.3 Capabilities

Definition 2.6 (Capability Set). The *capability set* of architecture A is:

$$\text{Cap}(A) = \{r \in R \mid A \text{ satisfies } r\}$$

Examples of capabilities:

- “Support horizontal scaling”
- “Provide type provenance”
- “Enable independent deployment”
- “Satisfy single source of truth for class definitions”

² : [L17](#), [L19](#)

- “Allow polyglot persistence”

Definition 2.7 (Capability Satisfaction). Architecture A *satisfies* requirement r (written $A \models r$) if there exists an execution trace in (S, T) that meets r ’s specification.

2.4 Leverage

Definition 2.8 (Leverage). The *leverage* of architecture A is:

$$L(A) = \frac{|\text{Cap}(A)|}{\text{DOF}(A)}$$

Special cases:

1. **Unbounded Leverage:** As $|\text{Cap}(A)|$ grows for fixed $\text{DOF}(A) = 1$, leverage grows without bound. This is the metaprogramming regime: a single source with an unbounded number of derived capabilities. In any fixed finite model, L is a positive rational; “ $L = \infty$ ” is a shorthand for this limit.
2. **Unit Leverage** ($L = 1$): Linear relationship (n capabilities from n DOF)
3. **Sublinear Leverage** ($L < 1$): Antipattern (more DOF than capabilities)

Example 2.9 (Leverage Calculations). • **SSOT:** $\text{DOF} = 1$, $\text{Cap} = \{F, \text{uses of } F\}$ where each new use adds a capability

$\Rightarrow L$ grows without bound as derivations accumulate

- **Scattered Code (n copies):** $\text{DOF} = n$, $\text{Cap} = \{F\}$
 $\Rightarrow L = 1/n$ (antipattern!)

- **Generic REST Endpoint:** $\text{DOF} = 1$, $\text{Cap} = \{\text{serve } n \text{ use cases}\}$
 $\Rightarrow L = n$

- **Specific Endpoints:** $\text{DOF} = n$, $\text{Cap} = \{\text{serve } n \text{ use cases}\}$
 $\Rightarrow L = 1$

Definition 2.10 (Architectural Dominance). Architecture A_1 *dominates* A_2 (written $A_1 \succeq A_2$) if:

1. $\text{Cap}(A_1) \supseteq \text{Cap}(A_2)$ (at least same capabilities)
2. $L(A_1) \geq L(A_2)$ (at least same leverage)

A_1 *strictly dominates* A_2 (written $A_1 \succ A_2$) if $A_1 \succeq A_2$ with at least one inequality strict.

2.5 Modification Complexity

Definition 2.11 (Modification Complexity). For requirement change δR , the *modification complexity* is:

$$M(A, \delta R) = \text{expected number of independent changes to implement } \delta R$$

Theorem 2.12 (Modification Bounded by DOF). *For all architectures A and requirement changes δR :*

$$M(A, \delta R) \leq \text{DOF}(A)$$

with equality when δR affects all components.

Proof. By Definition 2.3, DOF counts independent axes. A requirement change δR decomposes into sub-changes along individual axes; sub-changes along distinct axes are independent. Therefore the number of independent changes is at most the number of axes, which is $\text{DOF}(A)$. Equality holds when δR has a non-trivial projection onto every axis. ■

Example 2.13 (SSOT vs Scattered). Consider changing a structural fact F with n use sites:

- **SSOT:** $M = 1$ (change at source, derivations update automatically)
- **Scattered:** $M = n$ (must change each copy independently)

2.6 Formalization in Lean

All definitions in this section are formalized in `Leverage/Foundations.lean`:

- **Architecture:** Structure with components, state, transitions, requirements
- **Architecture.leverage:** Leverage metric
- **Architecture.dominates:** Dominance relation
- **compose_dof:** Proposition 2.4
- **modification_eq_dof:** Theorem 2.12

3 Probability Model

We derive the relationship between DOF and error probability from `AbstractClassSystem`'s axis independence theorem.

3.1 Error Independence from Axis Orthogonality

The independence of errors is not an axiom; it is a consequence of axis orthogonality proved in `AbstractClassSystem` [3].

Theorem 3.1 (Error Independence). *If axes $\{A_1, \dots, A_n\}$ are orthogonal (`AbstractClassSystem`, `Theorem minimal_complete_unique_orthogonal`), then errors along each axis are statistically independent.*

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Proof. By Definition 2.1, the state space is the product $S = \prod_{c \in C} S_c$. `AbstractClassSystem`'s orthogonality theorem (`minimal_complete_unique_orthogonal`) establishes that a minimal complete axis set partitions the state space into independent dimensions: axis A_i is a projection onto a factor of the product, and $A_i \perp A_j$ means the projection onto factor i carries no information about factor j .

An *error along axis A_i* is the event E_i that the value on factor i deviates from specification. Since the factors are independent dimensions of the product space, the probability measure factorizes:

$$P(E_i \mid \text{state of factors } \neq i) = P(E_i).$$

³ : L37 ⁴ : L23

This is the standard characterization of independence for product probability spaces: the marginal on factor i is independent of the marginal on any other factor. Therefore $P(E_i \cap E_j) = P(E_i) \cdot P(E_j)$ for all $i \neq j$, and by induction the n events $E_1, \dots, E_n$ are mutually independent. ■

Corollary 3.2 (DOF = Independent Error Sources). *DOF(A) = n implies n independent sources of error, each with probability p.*

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Proof. DOF counts independent axes (Section 2, Definition 2.3). By Theorem 3.1, independent axes have independent errors. ■

Theorem 3.3 (Error Compounding). *For a system to be correct, all n independent axes must be error-free. Errors compound multiplicatively.*

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Proof. By **Ssot**'s coherence theorem (**oracle_arbitrary**), incoherence in any axis violates system correctness. An error in axis A_i introduces incoherence along A_i . Therefore, correctness requires $\bigwedge_{i=1}^n \neg \text{error}(A_i)$. By Theorem 3.1, this probability is $(1 - p)^n$. ■

3.2 Error Probability Formula

Theorem 3.4 (Error Probability). *For architecture with n DOF and per-component error rate p:*

$$P_{\text{error}}(n) = 1 - (1 - p)^n$$

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Proof. By Theorem 3.1 (derived from **AbstractClassSystem**'s orthogonality), each DOF has independent error probability p , so each is correct with probability $(1 - p)$. By Theorem 3.3, all n DOF must be correct:

$$P_{\text{correct}}(n) = (1 - p)^n$$

Therefore:

$$P_{\text{error}}(n) = 1 - P_{\text{correct}}(n) = 1 - (1 - p)^n$$

■

Corollary 3.5 (Linear Approximation). *For fixed $p > 0$, the linear expected-error model and the exact series model induce the same ordering on architectures by DOF.*

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Proof. By Theorem 3.10, ordering equivalence is exact under the discrete model used in the mechanization. ■

Corollary 3.6 (DOF-Error Monotonicity). *For architectures A_1, A_2 :*

$$\text{DOF}(A_1) < \text{DOF}(A_2) \implies P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$$

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Proof. $P_{\text{error}}(n) = 1 - (1 - p)^n$ is strictly increasing in n for $p \in (0, 1)$. ■

⁵ : L15, L23 ⁶ : L20, L41 ⁷ : L24, L40 ⁸ : L16, L39 ⁹ : L33

3.3 Expected Errors

Theorem 3.7 (Expected Error Bound). *Expected number of errors in architecture A:*

$$\mathbb{E}[\# \text{ errors}] = p \cdot \text{DOF}(A)$$

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Proof. By linearity of expectation:

$$\mathbb{E}[\# \text{ errors}] = \sum_{i=1}^{\text{DOF}(A)} P(\text{error in DOF}_i) = \sum_{i=1}^{\text{DOF}(A)} p = p \cdot \text{DOF}(A)$$

■

Example 3.8 (Concrete Calculations). Assume $p = 0.01$ (1% per-component error rate):

- DOF = 1: $P_{\text{error}} = 1 - 0.99 = 0.01$ (1%)
- DOF = 10: $P_{\text{error}} = 1 - 0.99^{10} \approx 0.096$ (9.6%)
- DOF = 100: $P_{\text{error}} = 1 - 0.99^{100} \approx 0.634$ (63.4%)

3.4 Connection to Reliability Theory

The error model has a direct interpretation in classical reliability theory [1], connecting software architecture to a mature mathematical framework with 60+ years of theoretical development.

Theorem 3.9 (DOF-Reliability Isomorphism). *An architecture with $\text{DOF} = n$ is isomorphic to a series reliability system with n components. The isomorphism:*

1. **Preserves ordering:** If $\text{DOF}(A_1) < \text{DOF}(A_2)$, then $P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$
2. **Is invertible:** Round-trip mapping preserves DOF exactly
3. **Connects domains:** $P_{\text{error}}(n) = 1 - R_{\text{series}}(n)$ where $R_{\text{series}}(n) = (1 - p)^n$

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Interpretation: Each DOF is a “component” that must work correctly. This is the reliability analog of Theorem 3.1, which derives error independence from axis orthogonality.

Theorem 3.10 (Ordering Equivalence (Exact)). *For architectures A_1, A_2 with equal capabilities and error rate $p > 0$, the linear model and the exact series model induce the same DOF ordering:*

$$\text{DOF}(A_1) \leq \text{DOF}(A_2) \iff (\mathbb{E}[\text{errors}(A_1)] \cdot d) \leq (\mathbb{E}[\text{errors}(A_2)] \cdot d),$$

where d is the denominator of the discrete error-rate representation.

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Proof. This is theorem `ordering_equivalence_exact` in `Leverage/Probability.lean`. ■

Linear Approximation Justification: For small p (the software engineering regime where $p \approx 0.01$), the linear model $P_{\text{error}} \approx n \cdot p$ is:

1. Order-equivalent to the exact series model in the mechanized discrete representation
2. Monotone in DOF for fixed positive error rate
3. Cleanly formalized in natural-number arithmetic

¹⁰ : L25-26 ¹¹ : L22, L27 ¹² : L32, L39

3.5 Epistemic Grounding

The probability model is not axiomatic; it is derived from the epistemic foundations in `AbstractClassSystem` and `Ssot`:

1. `AbstractClassSystem` proves axis orthogonality (`minimal_complete_unique_orthogonal`)
2. **Theorem 3.1** derives error independence from orthogonality
3. `Ssot` establishes that $\text{DOF} = 1$ guarantees coherence (`oracle_arbitrary`)
4. **Theorem 3.3** connects errors to incoherence

This derivation chain ensures the probability model rests on proved foundations, not assumed axioms.

3.6 Leverage Gap: Quantitative Predictions

The leverage framework provides *quantitative*, empirically testable predictions about modification costs.

Theorem 3.11 (Leverage Gap). *For architectures with equal capabilities, the modification cost ratio equals the inverse leverage ratio:*

$$\frac{\text{ModCost}(A_2)}{\text{ModCost}(A_1)} = \frac{\text{DOF}(A_2)}{\text{DOF}(A_1)} = \frac{L(A_1)}{L(A_2)}$$

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Theorem 3.12 (Testable Prediction). *If architecture A_1 has $n\times$ lower DOF than A_2 (for equal capabilities), then A_1 requires $n\times$ fewer expected modifications. This is empirically testable against PR/commit data.*

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Corollary 3.13 (DOF Ratio Predicts Error Ratio). *The ratio of expected errors between two architectures equals the ratio of their DOF:*

$$\frac{\mathbb{E}[\text{errors}(A_2)]}{\mathbb{E}[\text{errors}(A_1)]} = \frac{\text{DOF}(A_2)}{\text{DOF}(A_1)}$$

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These theorems transform architectural intuitions into testable hypotheses. A claim that “Pattern X is $3\times$ better than Pattern Y” can be verified by comparing DOF and measuring modification frequency in real codebases.

3.7 Formalization

Formalized in `Leverage/Probability.lean`:

- `dof_reliability_isomorphism`: Theorem 3.9 (the central isomorphism)
- `isomorphism_preserves_failure_ordering`: Ordering preservation
- `isomorphism_roundtrip`: Invertibility proof

13 : L30 14 : L42 15 : L21

- `ordering_equivalence_exact`: Theorem 3.10 (exact ordering equivalence)
- `linear_model_preserves_ordering`: Linear ordering support
- `leverage_gap`: Theorem 3.11
- `testable_modification_prediction`: Theorem 3.12
- `dof_ratio_predicts_error_ratio`: Corollary 3.13
- `lower_dof_lower_errors`: Corollary 3.6
- `expected_errors_from_linearity`: Theorem 3.7
- `ssot_minimal_errors`: SSOT minimality

4 Main Theorems

We prove the core results connecting leverage to error probability and architectural optimality. The theoretical structure is:

1. **DOF-Reliability Isomorphism** (Theorem 3.9): Maps architecture to reliability theory
2. **Leverage-Error Tradeoff** (Theorem 4.2): Connects leverage to error probability
3. **Physical Edit-Energy Floor** (Theorem 4.3): DOF controls minimum edit-energy
4. **Optimality Criterion** (Theorem 4.8): Correctness in a fixed capability class
5. **Composition Stability** (Theorem 4.9): Composition preserves leverage lower bounds

4.1 Recap: DOF-Reliability Isomorphism

The core theoretical contribution (stated in Section 1.3) is that DOF maps formally to series system components. This enables all subsequent results by connecting software architecture to the mature mathematical framework of reliability theory.

Key properties of the isomorphism:

- **Preserves ordering**: If $\text{DOF}(A_1) < \text{DOF}(A_2)$, then $P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$
- **Invertible**: An architecture can be reconstructed from its series system representation
- **Compositional**: $\text{DOF}(A_1 \oplus A_2) = \text{DOF}(A_1) + \text{DOF}(A_2)$ (series systems combine)

4.2 The Leverage Maximization Principle

Given the DOF-Reliability Isomorphism, the following is a *corollary*:

Theorem 4.1 (Leverage Maximization Principle). *For any architectural decision with alternatives $A_1, \dots, A_n$ meeting capability requirements, the optimal choice maximizes leverage:*

$$A^* = \arg \max_{A_i} L(A_i) = \arg \max_{A_i} \frac{|\text{Capabilities}(A_i)|}{\text{DOF}(A_i)}$$

¹⁶

Note: This is *not* the central theorem; it is a consequence of the DOF-Reliability Isomorphism. The isomorphism is the deep result; “maximize leverage” is the actionable heuristic derived from it.

¹⁶ : L28, L31

4.3 Leverage-Error Tradeoff

Theorem 4.2 (Leverage-Error Tradeoff). *For architectures A_1, A_2 with equal capabilities:*

$$\text{Cap}(A_1) = \text{Cap}(A_2) \wedge L(A_1) > L(A_2) \implies P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$$

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Proof. Given: $\text{Cap}(A_1) = \text{Cap}(A_2)$ and $L(A_1) > L(A_2)$.

Since $L(A) = |\text{Cap}(A)|/\text{DOF}(A)$ and capabilities are equal:

$$\frac{|\text{Cap}(A_1)|}{\text{DOF}(A_1)} > \frac{|\text{Cap}(A_2)|}{\text{DOF}(A_2)}$$

With $|\text{Cap}(A_1)| = |\text{Cap}(A_2)|$:

$$\frac{1}{\text{DOF}(A_1)} > \frac{1}{\text{DOF}(A_2)} \implies \text{DOF}(A_1) < \text{DOF}(A_2)$$

By Corollary 3.6:

$$\text{DOF}(A_1) < \text{DOF}(A_2) \implies P_{\text{error}}(A_1) < P_{\text{error}}(A_2)$$

■

Corollary: Maximizing leverage minimizes error probability (for fixed capabilities).

4.4 Physical Edit-Energy Floor

Theorem 4.3 (Physical Edit-Energy Floor). *Fix a physical edit model $\mathcal{M}$ with per-edit conversion constant $j_{\mathcal{M}} > 0$. For any architecture A :*

$$E_{\min}(A; \mathcal{M}) = j_{\mathcal{M}} \cdot \text{DOF}(A).$$

Hence $\text{DOF}(A_1) < \text{DOF}(A_2)$ implies $E_{\min}(A_1; \mathcal{M}) < E_{\min}(A_2; \mathcal{M})$.

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Proof. In the mechanized model, modification complexity is definitionally DOF. Multiplying by a positive per-edit conversion constant gives the lower bound and strict monotonicity in DOF. ■

Corollary 4.4 (Leverage-Energy Monotonicity in a Capability Class). *For architectures A_1, A_2 with equal capabilities, if $L(A_1) > L(A_2)$ then*

$$E_{\min}(A_1; \mathcal{M}) < E_{\min}(A_2; \mathcal{M}),$$

and the induced energy gap is strictly positive.

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Theorem 4.5 (Budget Feasibility Boundary). *For physical model $\mathcal{M}$ and budget B , implementation feasibility is equivalent to clearing the floor:*

$$E_{\min}(A; \mathcal{M}) \leq B \iff \text{feasible}(A, B, \mathcal{M}).$$

Hence $B < E_{\min}(A; \mathcal{M})$ implies infeasibility. Also, for fixed B and equal capabilities, if $L(A_1) > L(A_2)$ and A_2 is feasible under B , then A_1 is feasible under B .

¹⁷ : L29 ¹⁸ : L1, L5 ¹⁹ : L3, L6

Corollary 4.6 (Physical Assumption Necessity Witnesses). *The physical layer uses two explicit premises for no-go style infeasibility claims: positive per-edit conversion and an external budget bound. Dropping positivity admits a zero-floor countermodel. Dropping the external budget-bound premise blocks infeasibility conclusions because a feasible budget witness always exists.*

4.5 Metaprogramming Dominance

Theorem 4.7 (Metaprogramming Dominance). *For any $N \in \mathbb{N}$, there exists a metaprogramming architecture M with $\text{DOF}(M) = 1$ and $L(M) \geq N$.*

Proof. Let M be a metaprogramming architecture with a single source definition ($\text{DOF}(M) = 1$) and at least N derived capabilities. Then $L(M) = |\text{Cap}(M)|/1 \geq N$.

Modeling note. In any fixed finite model, $L(M)$ is a positive rational. The colloquial claim “leverage = ∞ ” means: for any bound N , the architecture can be extended to achieve $L \geq N$ while keeping $\text{DOF} = 1$. The Lean formalization proves the for-all- N version (L36). ■

4.6 Architectural Decision Criterion

Theorem 4.8 (Optimal Architecture). *Given requirements R , architecture A^* is optimal in its capability class if and only if:*

1. $\text{Cap}(A^*) \supseteq R$ (feasibility)
2. $\forall A'$ with $\text{Cap}(A') = \text{Cap}(A^*)$ and $\text{Cap}(A') \supseteq R$: $L(A^*) \geq L(A')$ (maximality in fixed capability class)

Proof. ($\Leftarrow$) Suppose A^* satisfies (1) and (2). Then A^* is feasible and has maximum leverage among feasible architectures in the same capability class. By Theorem 4.2, this minimizes error probability in that class.

($\Rightarrow$) If (1) fails, A^* is infeasible. If (2) fails, there exists A' in the same capability class with $L(A') > L(A^*)$, so $P_{\text{error}}(A') < P_{\text{error}}(A^*)$ by Theorem 4.2, contradicting optimality. ■

Decision Procedure:

1. Enumerate candidate architectures $\{A_1, \dots, A_n\}$
2. Filter: Keep only A_i with $\text{Cap}(A_i) \supseteq R$
3. Optimize: Choose $A^* = \arg \max_i L(A_i)$

²⁰ : L2, L4 ²¹ : L7-8 ²² : L35-36 ²³ : L34, L38

4.7 Leverage Composition

Theorem 4.9 (Leverage Composition). *For modular architecture $A = A_1 \oplus A_2$ with disjoint components:*

1. $DOF(A) = DOF(A_1) + DOF(A_2)$
2. if $L(A_1) \geq 1$ and $L(A_2) \geq 1$, then $L(A) \geq 1$

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Proof. (1) By Proposition 2.4.

(2) If $L(A_i) \geq 1$, then $|\text{Cap}(A_i)| \geq DOF(A_i)$ for $i = 1, 2$. Summing both inequalities gives

$$|\text{Cap}(A_1)| + |\text{Cap}(A_2)| \geq DOF(A_1) + DOF(A_2),$$

which is exactly $L(A) \geq 1$ under additive composition. ■

Interpretation: Composition preserves a baseline leverage floor under the stated assumptions.

Remark 4.10 (Composition Breaks SSOT). Theorem 4.9 preserves leverage *floors*, but composing two SSOT architectures breaks SSOT: if $DOF(A_1) = DOF(A_2) = 1$, then $DOF(A_1 \oplus A_2) = 2$. This is formalized as `compose_breaks_ssot` in `Leverage/BridgeToDQ.lean`. The composition tax is unavoidable: distributed systems consisting of k independent SSOT components have $DOF = k$, placing them in the coNP-hard regime ($\text{srnk} = k > 1$) with mandatory thermodynamic cost per decision cycle.

4.8 Formalization

Main optimization theorems are formalized in `Leverage/Theorems.lean`; physical edit-energy theorems are formalized in `Leverage/Physical.lean`:

- L29: Theorem 4.2
- L1 and L5: Theorem 4.3
- L3 and L6: Corollary 4.4
- L2 and L4: Theorem 4.5
- L7 and L8: Corollary 4.6
- L35 and L36: Theorem 4.7
- L34 and L38: Theorem 4.8
- L18 and L19: Theorem 4.9
- `Leverage/Physical.lean`: physical edit-energy floor theorems

²⁴ : L18-19

4.9 Integration with Dependencies

The leverage framework provides the unifying theory for results proved in `AbstractClassSystem` and `Ssot`:

Theorem 4.11 (`AbstractClassSystem` as Leverage Instance). *Nominal typing dominance from `AbstractClassSystem` [3] is an instance of leverage maximization:*

- *Nominal typing adds 4 B-dependent capabilities over duck typing*
- *DOF remains fixed in the mechanized typing instance*
- *Therefore nominal typing has strictly higher leverage*

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Theorem 4.12 (`Ssot` as Leverage Instance). *SSOT dominance from `Ssot` [4] is an instance of leverage maximization:*

- *SSOT fixes DOF at 1 for a structural fact*
- *Non-SSOT replication yields $DOF = n$ for the same fact*
- *Therefore leverage improves by factor n*

26

These integration theorems are formalized in:

- `Leverage/Typing.lean`
- `Leverage/SSOT.lean`

5 Five-Way Equivalence

This section establishes the central result: five independent characterizations of the same architectural property ($DOF = 1$ / SSOT) derived from five distinct first-principles frameworks. All proofs are machine-checked in Lean 4.

5.1 Framework 1: Engineering Optimization (Leverage)

Theorem 5.1 (Maximum Leverage). For any architecture A with $\text{Cap}(A) > 0$, A achieves maximum leverage among architectures with equal capabilities if and only if $\text{DOF}(A) = 1$.

Formally: $L(A) = \max_{A': \text{Cap}(A') = \text{Cap}(A)} L(A') \iff \text{DOF}(A) = 1$.

An architecture has the highest possible capability-to-DOF ratio exactly when it has a single degree of freedom.

Proof.

- (Forward) If $\text{DOF}(A) = 1$, then $L(A) = |\text{Cap}(A)|/1 = |\text{Cap}(A)|$. Any A' with same capabilities has $\text{DOF}(A') \geq 1$, so $L(A') \leq |\text{Cap}(A)| = L(A)$.
- (Backward) If A has maximum leverage but $\text{DOF}(A) > 1$, construct A' with $\text{DOF}(A') = 1$ and same capabilities. Then $L(A') = |\text{Cap}(A)| > |\text{Cap}(A)|/\text{DOF}(A) = L(A)$, contradiction.

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²⁵ : [L13-14](#) ²⁶ : [L10-11](#) ²⁷ : [L28](#), [L31](#), [L48](#)

5.2 Framework 2: Architectural Epistemic Coherence (Ssot)

Theorem 5.2 (Coherence). An architecture satisfies the Single Source of Truth property if and only if $\text{DOF}(A) = 1$.

Formally: $\text{SSOT}(A) \iff \text{DOF}(A) = 1$.

Epistemic coherence (one source of truth per structural fact) is equivalent to having a single degree of freedom.

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This is the $\text{DOF} = 1$ characterization from the Ssot coherence theorem.

5.3 Framework 3: Information-Theoretic Structural Rank (DecisionQuotient)

Theorem 5.3 (DOF-Structural Rank Isomorphism). For any architecture A , let $\text{canonicalDP}(n)$ be the canonical decision problem with n boolean coordinates. Then:

$$\text{srnk}(\text{canonicalDP}(\text{DOF}(A))) = \text{DOF}(A)$$

The structural rank (number of relevant decision coordinates) equals the degrees of freedom. The canonical encoding uses: states as n boolean coordinates, actions as either querying a coordinate or falling back, and utilities that reward correct coordinate identification.

The canonical encoding is:

- States: $\text{Fin } n \rightarrow \text{Bool}$ (n binary coordinates)
- Actions: $\text{Fin } n \oplus \text{Unit}$ (query coordinate i , or fallback)
- Utility: $u(\text{inl } i, s) = 2$ if $s(i) = \text{true}$, else 0; $u(\text{inr } (), s) = 1$

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Corollary 5.4. $\text{DOF} = 1$ if and only if $\text{srnk} = 1$.

Formally: $\text{DOF}(A) = 1 \iff \text{srnk}(\text{canonicalDP}(\text{DOF}(A))) = 1$.

An architecture has a single degree of freedom exactly when its canonical decision problem has minimal structural rank.

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5.4 Framework 4: Computational Complexity (Tractability)

Theorem 5.5 (Tractability Boundary). For the canonical decision problem with n coordinates:

1. If $n = 1$ ($\text{srnk} = 1$), sufficiency checking is in P (polynomial time).
2. If $n > 1$ ($\text{srnk} > 1$), sufficiency checking is coNP-hard.

When the structural rank is 1, determining whether current information suffices for an optimal decision is computationally tractable. When the structural rank exceeds 1, the same problem is computationally intractable (coNP-hard).

²⁸ : Proved in Ssot ([ORA1](#)) ²⁹ : [L43](#), [L46](#), [L51](#) ³⁰ : Immediate from Theorem 5.3

31

This connects the architectural property ($\text{DOF} = 1$) to computational feasibility of decision-making.

Corollary 5.6. $\text{DOF} = 1$ (SSOT) is the unique architecture class for which sufficiency checking is tractable.

Among architectures, only those with a single degree of freedom admit tractable sufficiency checking.

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5.5 Framework 5: Thermodynamic Selection (Statistical Physics)

Theorem 5.7 (Thermodynamic Selection). Let M be a thermodynamic model with Landauer calibration. For any architecture A with $\text{DOF} > 1$:

1. There exist decision instances where sufficiency checking requires $\Omega(2^{\text{DOF}})$ logical operations.
2. Under the physical axioms of [2] (LP38): Landauer (EP1), nontrivial state space (EP4), finite signal speed (LP43), finite universe energy (LP44) — $P \neq NP$ is proved, hence $P \neq \text{coNP}$; no polynomial-time sufficiency certification exists.
3. **(Unconditional)** Each sufficiency-check cycle incurs mandatory positive energy cost $\geq j_{\mathcal{M}} \cdot \text{srnk}(A) > j_{\mathcal{M}}$, proved from Landauer calibration and the $\text{DOF} = \text{srnk}$ identity alone, with no complexity hypothesis.

For $\text{DOF} = 1$, the energy lower bound is $j_{\mathcal{M}}$ (the Landauer minimum).

Architectures with more than one degree of freedom necessarily incur thermodynamic costs per decision cycle. Point 3 holds unconditionally: the energy bound follows from Landauer and the $\text{DOF} = \text{srnk}$ identity (L43). Point 2 holds under the physical axioms of [2] (LP38), which prove $P \neq NP$, hence $P \neq \text{coNP}$.

Physical assumptions (point 3 only):

- Positive Landauer constant ($j_{\mathcal{M}} > 0$) — cf. Theorem 4.4 (L3, L6)
- Landauer calibration ($k_B T \ln 2$ joules per bit)

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Physical edit-energy floor from Section 4.3 is a special case: DOF controls minimum edit-energy, and higher leverage implies lower energy in a fixed capability class (Theorem 4.3, L1, L5).

5.6 The Five-Way Equivalence

Theorem 5.8 (Five-Way Equivalence). For any architecture A with $\text{Cap}(A) > 0$, the following are equivalent:

$$\text{DOF}(A) = 1 \iff \text{max leverage} \iff \text{SSOT}(A) \iff \text{srnk}(A) = 1 \iff \text{tractable sufficiency} \iff \text{minimum therm}$$

Five independent scientific frameworks—engineering optimization, epistemic coherence, information geometry, computational complexity, and statistical physics—all select the same architectural property: having exactly one degree of freedom.

³¹ : Proved in `DecisionQuotient` (L56, L53), imported via `Leverage/BridgeToDQ.lean` ³² : Combine Corollary 5.4 with Theorem 5.5; L47 ³³ : L49, L54–55

Domain	Characterization of $\text{DOF} = 1$
Engineering Optimization	Maximum leverage
Architectural Epistemic	Single Source of Truth
Information Geometry	Structural rank = 1
Computational Complexity	Tractable sufficiency checking
Statistical Physics	Minimum thermodynamic cost per cycle

Proof. The equivalence follows from Theorems 5.1-5.7:

1. $\text{DOF} = 1 \iff \text{max leverage}$: Theorem 5.1 ([L31](#), [L28](#))
2. $\text{DOF} = 1 \iff \text{SSOT}$: Theorem 5.2 [[4](#)]
3. $\text{DOF} = \text{srnk}$: Theorem 5.3, so $\text{DOF} = 1 \iff \text{srnk} = 1$
4. $\text{srnk} = 1 \iff \text{tractable sufficiency}$: Theorem 5.5 [[2](#)]. Forward direction ($\text{srnk} = 1 \rightarrow \text{poly-nomial}$) is unconditional. Backward direction ($\text{tractable} \rightarrow \text{srnk} = 1$) holds under $\text{P} \neq \text{coNP}$, which follows from the physical axioms of [[2](#)] ([LP38](#)).
5. $\text{DOF} > 1 \iff \text{above-minimum thermodynamic cost}$: Theorem 5.7, so $\text{DOF} = 1 \iff \text{minimum thermodynamic cost}$

The transitivity of logical equivalence gives the five-way equivalence.

[34](#)

Corollary 5.9 (Convergence). Within the `DecisionQuotient` canonical decision problem framework, the boolean-coordinate encoding (states as n boolean coordinates, actions as coordinate queries or fallback, utilities rewarding correct identification) is the unique decision structure with $n = 1$ that simultaneously satisfies all five characterizations.

In the canonical decision problem representation, there is exactly one $n = 1$ structure, and all five characterizations select it. This is a consequence of the equivalences, not an additional theorem. Note: there are many architectures with $\text{DOF} = 1$ (monolith, single DB table, single endpoint); the uniqueness claim is scoped to the canonical decision problem encoding.

Proof. Each characterization is equivalent to $\text{DOF} = 1$ (Theorems 5.1–5.7). Within the canonical encoding, $\text{DOF} = 1$ fixes $n = 1$, and $\text{Fin } 1 \rightarrow \text{Bool}$ has a unique element. All five requirements therefore select the same structure.

[35](#)

5.7 Formalization

The following Lean 4 proofs establish the connections:

- `Leverage/Foundations.lean`:
 - [L31](#), [L28](#) ([L50](#)): $\text{DOF} = 1 \rightarrow \text{max leverage}$
 - [L48](#): $\text{max leverage} \rightarrow \text{DOF} = 1$
 - [L44](#): biconditional

³⁴ : [L43-44](#), [L47](#), [L55](#), [ORA1](#) ³⁵ : All five requirements force $n = 1$ by the above theorems; uniqueness within the canonical encoding follows from $|\text{Fin } 1 \rightarrow \text{Bool}| = 2$

- `Leverage/BridgeToDQ.lean`:
 - [L43](#): $\text{DOF} = \text{srank}$
 - [L51](#): $\text{DOF} = 1 \rightarrow \text{srank} = 1$
 - [L46](#): $\text{DOF} > 1 \rightarrow \text{srank} > 1$
 - [L54](#): $\text{DOF} > 1 \rightarrow \text{mandatory energy}$
 - [L47](#): $\text{max leverage} \rightarrow \text{tractable}$
- `DecisionQuotient/Tractability/StructuralRank.lean`: $\text{srank} = 1 \rightarrow \text{P}$ ([L56](#)), $\text{srank} > 1 \rightarrow \text{coNP-hard}$ ([L53](#))
- `DecisionQuotient/ThermodynamicLift.lean`: energy lower bounds under Landauer calibration ([L49](#))

The cross-module dependency chain is live: `Leverage` $\rightarrow$ `DecisionQuotient` $\rightarrow$ `Mathlib`.

5.8 Relation to Prior Sections

This section subsumes and extends Section 4’s results:

- Theorem 4.1 (Leverage-Error Tradeoff, [L29](#)) is a consequence of the equivalence between leverage and DOF
- Theorem 4.2 (Modification Complexity Gap, [L30](#)) follows from $\text{DOF} = 1$ minimizing modification cost
- Theorem 4.3 (Optimal Architecture, [L38](#), [L34](#)) is strengthened: the optimal architecture is now characterized by five independent properties
- Theorem 4.4 (Metaprogramming Dominance, [L36](#), [L35](#)) is a special case: unbounded derivations from a single source achieve $\text{DOF} = 1$

5.9 England Replication Inequality (Proved)

All previously open conjectures are now proved. The England Replication Inequality is mechanized in `Leverage/BridgeToDQ.lean` ([L45](#)).

Theorem 5.1 (England Replication Inequality). *Let $\Delta S_{\min}(\text{srank}) = \text{srank} \cdot k_B \ln 2$ be the minimal entropy production under Landauer calibration. For a single-source architecture ($\text{srank} = 1$) and a k -copy replication architecture ($\text{srank} = k$):*

$$\Delta S_{\min}(1) + k_B \ln k \leq \Delta S_{\min}(k)$$

equivalently, $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$.

³⁶

Proof. The gap is $(k - 1) \cdot k_B \ln 2$. Since $k \leq 2^{k-1}$ ([L52](#)), taking logs gives $\ln k \leq (k - 1) \ln 2$, so the gap is $\geq k_B \ln k$. ■

Modeling note. $\Delta S_{\min}$ is a definition within the model: the exact Landauer entropy cost per cycle, not a lower bound on arbitrary implementations. The “min” refers to physical optimality outside the formalism.

³⁶ : [L45](#)

6 Related Work

6.1 Landauer and Thermodynamic Information Processing

Landauer’s principle provides the calibration from logical discrimination to minimum entropy production and energy cost. Our use of Landauer is deliberately finite and structural: the relevant parameter is not a Shannon rate or an asymptotic code family, but the number of independent coordinates that must be resolved in a sufficiency check. In that sense the paper is closer to a mechanized finite-cost selection theorem than to a phenomenological thermodynamic analogy.

The England-style replication inequality proved here sits in the same non-equilibrium neighborhood, but its derivation is intentionally discrete. The entropy gap is reduced to counting over a finite state family and the elementary inequality $k \leq 2^{k-1}$; the physics enters through Landauer calibration rather than through a separate stochastic-process ansatz.

6.2 Zero-Error and Structural Information Theory

The structural-rank side of the paper depends directly on the companion decision-quotient theory, where sufficiency checking, rank collapse, and hardness boundaries are developed in full. The present paper does not reprove that theory; instead it identifies the coordinate along which the thermodynamic lift should be read. This places the argument in the orbit of finite zero-error and decision-relevant information theory rather than classical average-case coding.

The rank-1 regime is special because it is simultaneously the single surviving decision coordinate, the tractable sufficiency regime, and the minimum calibrated decision-cost regime. That convergence is the conceptual novelty of the present paper relative to the companion structural-rank manuscript.

6.3 Reliability and Discrete Failure Composition

The reliability interpretation of DOF is used here as a discrete bridge: an n -degree-of-freedom system behaves like a series system with n independent failure points. We use that bridge not as the main application domain but as a mathematically transparent representation of why multiplicative error composition and additive decision cost are both controlled by the same structural parameter.

6.4 Formal Verification and Mechanized Cross-Paper Theory

All main statements are machine-checked in Lean 4, and the cross-paper dependency chain is live rather than informal. The mechanization matters here because the paper depends on results transported across distinct theorem families: orthogonality from `AbstractClassSystem`, coherence from `Ssot`, and rank/tractability/thermodynamic lift from `DecisionQuotient`. The present manuscript packages those results into a single equivalence theorem with an auditable proof trail.

7 Conclusion

7.1 Summary

The central result is the Five-Way Equivalence (Theorem 5.8): five independent scientific frameworks all characterize the single-source condition ($\text{DOF} = 1$).

Framework	DOF = 1 means	Source
Engineering optimization	Maximum leverage $L = \text{Cap} /\text{DOF}$	Leverage
Epistemic coherence	Single Source of Truth	Ssot
Information geometry	Structural rank = 1	DecisionQuotient
Computational complexity	Tractable sufficiency checking	DecisionQuotient
Statistical physics	Minimum thermodynamic cost per cycle	Leverage, DecisionQuotient

Reliability-style and edit-complexity statements are corollaries of the same underlying parameter, but they are not the main point of this version. The main point is that structural rank, tractability, and calibrated thermodynamic cost converge on the same discrete regime. All of these statements are machine-checked in Lean 4 with live cross-module imports (`Leverage` $\rightarrow$ `DecisionQuotient` $\rightarrow$ `Ssot` $\rightarrow$ `Mathlib`).

What is new in Leverage relative to `AbstractClassSystem` and `Ssot`:

- The convergence theorem itself: `AbstractClassSystem` and `Ssot` contain two of the five equivalences; this paper closes the chain.
- The thermodynamic selection theorem ([L55](#) in `BridgeToDQ.lean`): mandatory energy under Landauer calibration for $\text{DOF} > 1$, proved unconditionally.
- The identification of structural rank as the information coordinate for DOF.
- The England Replication Inequality ([L45](#)): $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$, giving the thermodynamic advantage of SSOT a quantitative non-equilibrium interpretation. No open conjectures remain.

7.2 No Open Conjectures

All conjectures from the original submission are now proved in `Leverage/BridgeToDQ.lean`:

- [L55](#): unconditional energy bound, no $P \neq \text{coNP}$ hypothesis.
- [L49](#): quantitative Landauer-linear energy bound.
- [L45](#): $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$ (Theorem [5.1](#)).

7.3 Limitations

1. Independence Assumption: The probability model treats DOF-level errors as independent under the orthogonality assumptions from `AbstractClassSystem` [\[3\]](#). Real systems may have correlated failure modes not captured by the isomorphism.

2. Constant Error Rate: Assumes p is uniform across components. Some components are more error-prone than others; a weighted version is future work.

3. $P \neq \text{coNP}$ Hypothesis (resolved): The thermodynamic energy bound in Theorem 5.7, point 3 is now proved unconditionally ([L55](#)). Point 2 (complexity hardness) holds under $P \neq \text{coNP}$, which follows from the physical axioms of [\[2\]](#) ([LP38](#)).

4. Capability Quantification: The leverage side still uses qualitative capability counting. That is adequate for the equivalence theorem but not yet the final word for weighted physical or engineering valuations.

7.4 Impact

For mathematical physics: the England Replication Inequality (Theorem 5.1) is reduced to finite counting plus Landauer calibration. The result gives a mechanized, non-asymptotic entropy gap for replication without appealing to an informal thermodynamic analogy.

For information theory and complexity: the paper identifies structural rank as the information coordinate of DOF and shows that the minimum-cost thermodynamic regime coincides with the tractable sufficiency regime.

For formalized theory building: the argument is an example of cross-paper theorem transport in Lean: a structural parameter introduced in one theorem family is lifted into complexity and physics in another, and the full equivalence remains auditable at the proof-artifact level.

7.5 Final Remarks

This paper proves finite thermodynamic and structural consequences of the single-source condition from first principles. The regime $\text{DOF} = 1$ is not merely a software heuristic; it is the unique point at which leverage, coherence, structural rank, tractable sufficiency, and minimum calibrated decision cost coincide.

All theorems are machine-checked. The thermodynamic lower bound is unconditional, and the replication inequality gives the entropy premium for multiplicity as $k_B \ln k$ per cycle. The companion software-engineering split develops the architectural and case-study implications separately; the present manuscript isolates the mathematical-physics core.

A Lean Proof Artifacts

This appendix reports machine-check status and proof traceability directly from source and generated mapping artifacts.

A.1 Verification Status

Lean summary: 75943 lines, 3301 theorems/lemmas, 0 sorry, across 280 files.

File	Lines	Theorems/Lemmas
LambdaDR.lean	343	24
Leverage.lean	115	0
Leverage/Foundations.lean	234	13
Leverage/Probability.lean	841	39
Leverage/Theorems.lean	303	20
Leverage/SSOT.lean	192	13
Leverage/Typing.lean	210	15
Leverage/Examples.lean	184	4
Leverage/WeightedLeverage.lean	348	13
Leverage/BridgeToDQ.lean	433	26
Leverage/FiveWayEquivalence.lean	149	7
Leverage/CrossPaperDependencies.lean	332	22
lakefile.lean	21	0
Total	75943	3301

Build command: `cd proofs && lake build`

A.2 Claim Coverage Matrix

Paper claim	Lean handle
Corollary 3.2: DOF = Independent Error Sources	L15 , L23
Corollary 3.6: DOF-Error Monotonicity	L33
Corollary 3.13: DOF Ratio Predicts Error Ratio	L21 , L42
Corollary 4.4: Leverage-Energy Monotonicity in a Capability Class	L3 , L6
Corollary 3.5: Linear Approximation	L16 , L39
Corollary 4.6: Physical Assumption Necessity Witnesses	L2 , L8 , L7 , L4
Definition 2.4: Architecture	L17 , L19
Theorem 3.10: Ordering Equivalence (Exact)	L32 , L39
Theorem 4.9: Leverage Composition	L18 , L19
Theorem 3.9: DOF-Reliability Isomorphism	L22 , L27
Theorem 5.1: England Replication Inequality	L45
Theorem 3.3: Error Compounding	L20 , L41
Theorem 3.1: Error Independence	L23
Theorem 3.4: Error Probability	L24 , L40
Theorem 3.7: Expected Error Bound	L25 , L26
Theorem 1.1: Five-Way Equivalence	<i>(no derived Lean handle found)</i>
Theorem 4.2: Leverage-Error Tradeoff	L29
Theorem 3.11: Leverage Gap	L30
Theorem 4.1: Leverage Maximization Principle	L28 , L31
Theorem 4.7: Metaprogramming Dominance	L35 , L36
Definition 2.12: Capability Set	L37
Unbounded Advantage	L13
Theorem 4.8: Optimal Architecture	L34 , L38
Theorem 4.11: <code>AbstractClassSystem</code> as Leverage Instance	L13 , L14
Theorem 4.12: <code>Ssot</code> as Leverage Instance	L10 , L11 , L13 , L14
Theorem 4.5: Budget Feasibility Boundary	L2 , L3 , L4 , L6
Theorem 4.3: Physical Edit-Energy Floor	L5 , L1
SSOT Architecture	L11
Theorem 3.12: Testable Prediction	L30 , L42

Notes: (1) Full rows come from theorem-local inline anchors in this paper. (2) Derived rows are filled by dependency/scaffold claim-handle derivation (same paper-handle label across proof dependencies). (3) Unmapped means no local anchor and no derivable dependency support were found. Auto summary: mapped 28/29 (full=28, derived=0, unmapped=1).

A.3 Lean Handle Map

ID	Lean Handle / Source	ID	Lean Handle / Source
BA7	Physics.BoundedAcquisition.energy_ge_srank_cost paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	DQ1	DecisionQuotient.Physics.BoundedAcquisition. energy_ge_srank_cost paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
EP1	Physics.LocalityPhysics.landauer_principle	EP4	Physics.LocalityPhysics.nontrivial_physics paper4/DecisionQuotient/Physics/ LocalityPhysics.lean
FXI1	fixed_axis_incompleteness paper1/axis_framework.lean	L1	matroid_basis_equicardinality paper1/axis_framework.lean
L2	Leverage.Physical.feasible_iff_floor_le_budget Leverage/Physical.lean	L3	Leverage.Physical.higher_leverage_same_caps_ implies_lower_energy Leverage/Physical.lean
L4	l4_exchange_wrapper paper1/HandleAliases.lean	L5	l5_exchange_wrapper paper1/HandleAliases.lean
L6	nonorthogonal_complete_has_redundant_axis paper1/axis_framework.lean	L7	exists_semanticallyMinimal_subset paper1/axis_framework.lean
L8	exists_orthogonal_semanticallyMinimal_subset paper1/axis_framework.lean	L10	Leverage.SSOT.paper2_is_leverage_instance Leverage/SSOT.lean
L11	Leverage.SSOT.ssot_leverage_dominance Leverage/SSOT.lean	L13	Leverage.Typing.nominal_dominates_duck Leverage/Typing.lean
L14	Leverage.Typing.paper1_is_leverage_instance Leverage/Typing.lean	L15	Leverage.architecture_axes_independent Leverage/Probability.lean
L16	Leverage.bernoulli_justifies_linear_model Leverage/Probability.lean	L17	Leverage.compose_dof Leverage/Foundations.lean
L18	Leverage.composition_caps_additive Leverage/Theorems.lean	L19	Leverage.composition_dof_additive Leverage/Theorems.lean
L20	Leverage.correctness_probability Leverage/Probability.lean	L21	Leverage.dof_ratio_predicts_error_ratio Leverage/Probability.lean
L22	Leverage.dof_reliability_isomorphism Leverage/Probability.lean	L23	Leverage.error_independence_from_orthogonality Leverage/Probability.lean
L24	Leverage.error_probability_denom_pos Leverage/Probability.lean	L25	Leverage.expected_errors_from_linearity Leverage/Probability.lean
L26	Leverage.expected_errors_linear Leverage/Probability.lean	L27	Leverage.isomorphism_preserves_failure_ ordering Leverage/Probability.lean
L28	Leverage.leverage_caps_principle Leverage/Theorems.lean	L29	Leverage.leverage_error_tradeoff Leverage/Theorems.lean
L30	Leverage.leverage_gap Leverage/Probability.lean	L31	Leverage.leverage_maximization_principle Leverage/Theorems.lean
L32	Leverage.linear_model_preserves_ordering Leverage/Probability.lean	L33	Leverage.lower_dof_lower_errors Leverage/Probability.lean
L34	Leverage.max_leverage_is_optimal Leverage/Theorems.lean	L35	Leverage.metaprogramming_dominates Leverage/Theorems.lean
L36	Leverage.metaprogramming_unbounded_leverage Leverage/Theorems.lean	L37	Leverage.modification_eq_dof Leverage/Foundations.lean
L38	Leverage.optimal_minimizes_error Leverage/Theorems.lean	L39	Leverage.ordering_equivalence_exact Leverage/Probability.lean
L40	Leverage.series_error_probability Leverage/Probability.lean	L41	Leverage.system_is_correct Leverage/Probability.lean
L42	Leverage.testable_modification_prediction Leverage/Probability.lean	L43	dof_eq_srank Leverage/BridgeToDQ.lean
L44	dof_one_iff_max_leverage Leverage/Foundations.lean	L45	england_replication_inequality Leverage/BridgeToDQ.lean
L46	incoherent_srank_gt_one Leverage/BridgeToDQ.lean	L47	max_coherence_forces_tractability Leverage/BridgeToDQ.lean
L48	max_leverage_forces_dof_one Leverage/Foundations.lean	L49	srank_energy_lower_bound Leverage/BridgeToDQ.lean
L50	ssot_max_leverage Leverage/Foundations.lean	L51	ssot_srank_one Leverage/BridgeToDQ.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
L52	succ_le_two_pow Leverage/BridgeToDQ.lean	L53	sufficiency_conp_hard
L54	thermodynamic_selection Leverage/BridgeToDQ.lean	L55	thermodynamic_selection_unconditional Leverage/BridgeToDQ.lean
L56	tractable_bounded_core paper4/DecisionQuotient/ClaimClosure.lean	L57	Leverage.SSOT.modification_ratio Leverage/SSOT.lean
LP38	Physics.LocalityPhysics.pne_np_necessary_for_ physics paper4/DecisionQuotient/Physics/ LocalityPhysics.lean	LP43	Physics.LocalityPhysics.without_separation_no_ independence paper4/DecisionQuotient/Physics/ LocalityPhysics.lean
LP44	Physics.LocalityPhysics.without_finite_ capacity_no_gap paper4/DecisionQuotient/Physics/ LocalityPhysics.lean	ORA1	oracle_arbitrary paper2/Ssot/Coherence.lean

A.4 Proof Hardness Index

Paper handle	Hardness profile	Regime tags	Lean support
cor:dof-errors	unspecified	-	L15, L23
cor:dof-monotone	unspecified	-	L33
cor:dof-ratio	unspecified	-	L21, L42
cor:leverage-energy	unspecified	-	L3, L6
cor:linear-approx	unspecified	-	L16, L39
cor:physical-	unspecified	-	L2, L8, L7, L4
assumption-necessity			
prop:dof-additive	unspecified	-	L17, L19
thm:approx-bound	unspecified	-	L32, L39
thm:composition	unspecified	-	L18, L19
thm:dof-reliability	unspecified	-	L22, L27
thm:england	unspecified	-	L45
thm:error-compound	unspecified	-	L20, L41
thm:	unspecified	-	L23
error-independence			
thm:error-prob	unspecified	-	L24, L40
thm:expected-errors	unspecified	-	L25, L26
thm:five-way	unspecified	-	(no derived Lean handle found)
thm:leverage-error	unspecified	-	L29
thm:leverage-gap	unspecified	-	L30
thm:leverage-max	unspecified	-	L28, L31
thm:metaprog	unspecified	-	L35, L36
thm:mod-bound	unspecified	-	L37
thm:nominal-leverage	unspecified	-	L13
thm:optimal	unspecified	-	L34, L38
thm:	unspecified	-	L13, L14
paper1-integration			
thm:	unspecified	-	L10, L11, L13, L14
paper2-integration			
thm:physical-budget-	unspecified	-	L2, L3, L4, L6
boundary			
thm:physical-energy-	unspecified	-	L5, L1
floor			
thm:ssot-leverage	unspecified	-	L11
thm:	unspecified	-	L30, L42
testable-prediction			

Auto summary: indexed 29 claims by hardness profile (unspecified=29).

B Notes on assumptions and extensions

This appendix lists the principal modeling assumptions and common extensions relevant when applying the leverage framework:

- **Independence and orthogonality:** Error-independence at the DOF level is derived from axis orthogonality assumptions; when dependencies are present, leverage remains a useful comparative metric but probabilistic models should be adjusted to account for correlation.
- **Capability counting:** The core theorems require only relative ordering of capabilities; cardinality serves as a practical proxy for capability breadth in examples and case studies.
- **Multi-objective concerns:** Leverage addresses error probability specifically. Performance, security, and other attributes require multi-objective analysis (e.g., Pareto frontiers) before operational decisions.
- **Implementations:** SSOT and other instantiations depend on language and platform features; the theoretical principle is implementation-agnostic.
- **Future work:** Extensions include explicit correlated-error models, weighted capability measures, and integration into multi-criteria architectural decision frameworks.

C Complete Theorem Index

Paper-level labeled claims in this manuscript:

Foundations (Section 2):

- Proposition [2.4](#) (DOF Additivity)
- Theorem [2.12](#) (Modification Bounded by DOF)

Probability Model (Section 3):

- Theorem [3.1](#)
- Corollary [3.2](#)
- Theorem [3.3](#)
- Theorem [3.4](#)
- Corollary [3.5](#)
- Corollary [3.6](#)
- Theorem [3.7](#)
- Theorem [3.9](#)
- Theorem [3.10](#)
- Theorem [3.11](#)
- Theorem [3.12](#)

- Corollary [3.13](#)

Main Results (Section 4):

- Theorem [4.1](#)
- Theorem [4.2](#)
- Theorem [4.3](#)
- Corollary [4.4](#)
- Theorem [4.5](#)
- Corollary [4.6](#)
- Theorem [4.7](#)
- Theorem [4.8](#)
- Theorem [4.9](#)
- Theorem [4.11](#)
- Theorem [4.12](#)

Instances (Section 5):

- Theorem ??
- Theorem ??

Mechanization status: 75943 lines, 3301 theorems/lemmas, 0 sorry, 280 files.

Primary Lean sources:

- `Leverage/Foundations.lean`
- `Leverage/Probability.lean`
- `Leverage/Theorems.lean`
- `Leverage/Physical.lean`
- `Leverage/SSOT.lean`
- `Leverage/Typing.lean`
- `Leverage/Examples.lean`
- `Leverage/WeightedLeverage.lean`
- `LambdaDR.lean`
- `Leverage.lean`

References

- [1] David A. Patterson and John L. Hennessy. *Computer Organization and Design: The Hardware/Software Interface*. Morgan Kaufmann, 2013. Discusses RAID and reliability formulas.
- [2] Tristan Simas. Computational complexity of physical counting. Zenodo, 2025. Companion paper. Proves coNP-hardness of sufficiency checking, tractable subcases, and $P \neq NP$ from physical axioms (Landauer, finite energy, finite signal speed, nontrivial state space). All results machine-checked in Lean 4.
- [3] Tristan Simas. Identification capacity and rate-query tradeoffs in classification systems. Zenodo, 2025. Paper 1 of this series. Proves axis orthogonality theorem.
- [4] Tristan Simas. Zero-incoherence capacity of interactive encoding systems: Achievability, converse, and side information bounds. Zenodo, 2025. Paper 2 of this series. Proves coherence impossibility theorems for programming languages.